Wellness Grocery Inventory Analysis - Clustering Pipeline Report

Course: ITCS 3162 - Introduction to Data Mining
Project: Wellness Grocery Inventory Analysis
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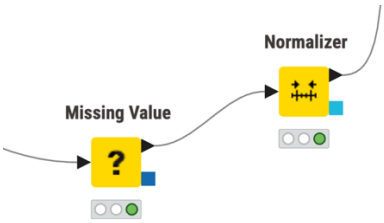
1. Data Preparation
The dataset was prepared by selecting the relevant numerical nutritional features and removing unnecessary columns. Missing values were handled using mean imputation to ensure the dataset was complete before modeling.

After cleaning, the data was normalized using standard scaling so that all features contributed equally to the clustering process. This step was necessary because variables such as kilocalories, fat, and water are measured on different scales. Without normalization, features with larger magnitudes would dominate the clustering results.



2. Dimensionality Reduction (PCA)
Principal Component Analysis (PCA) was applied to reduce the dimensionality of the dataset while preserving the most important variance. PCA transforms the original nutritional features into a smaller set of components that capture the underlying structure of the data.

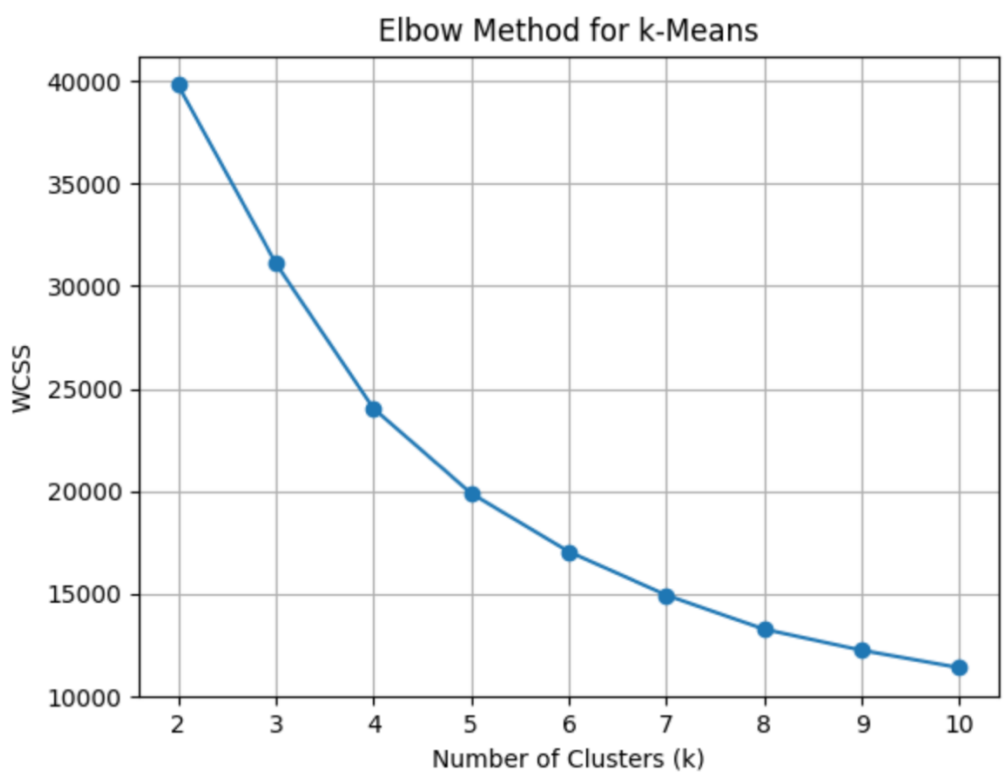
The first two principal components (PC0 and PC1) were used for visualization, allowing the dataset to be represented in a two-dimensional space while maintaining meaningful relationships between products.

	PCA dimension 0 Number (Float)	PCA dimension 1 Number (Float)
7	5.7	5.808
7	5.656	5.75
	7.39	7.057
2	1.408	2.523
9	1.522	2.603
8	1.137	2.44
1	0.789	2.166
	1.547	2.635

3. Elbow Method (Selecting k)
To determine the optimal number of clusters, the elbow method was implemented using Python to ensure efficient computation and clear visualization of WCSS trends. The within-cluster sum of squares (WCSS) was computed and plotted to observe how clustering error decreases as the number of clusters increases.

The elbow curve was generated using Python (program attached to GitHub) as a supporting tool to efficiently evaluate multiple k values. This approach allowed for a clear visualization of how clustering performance improves as k increases.

The resulting curve showed a clear bend around k = 5, indicating diminishing returns beyond this point. Therefore, k = 5 was selected as the optimal number of clusters and used in the KNIME workflow for clustering and analysis.



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4. k-Means Clustering
Using k = 5, the k-Means algorithm was applied to group products based on their nutritional features. Each product was assigned to one of five clusters representing different nutritional profiles within the dataset.

These clusters allow the grocery inventory to be segmented into meaningful groups based on patterns such as fat content, energy density, and carbohydrate levels.

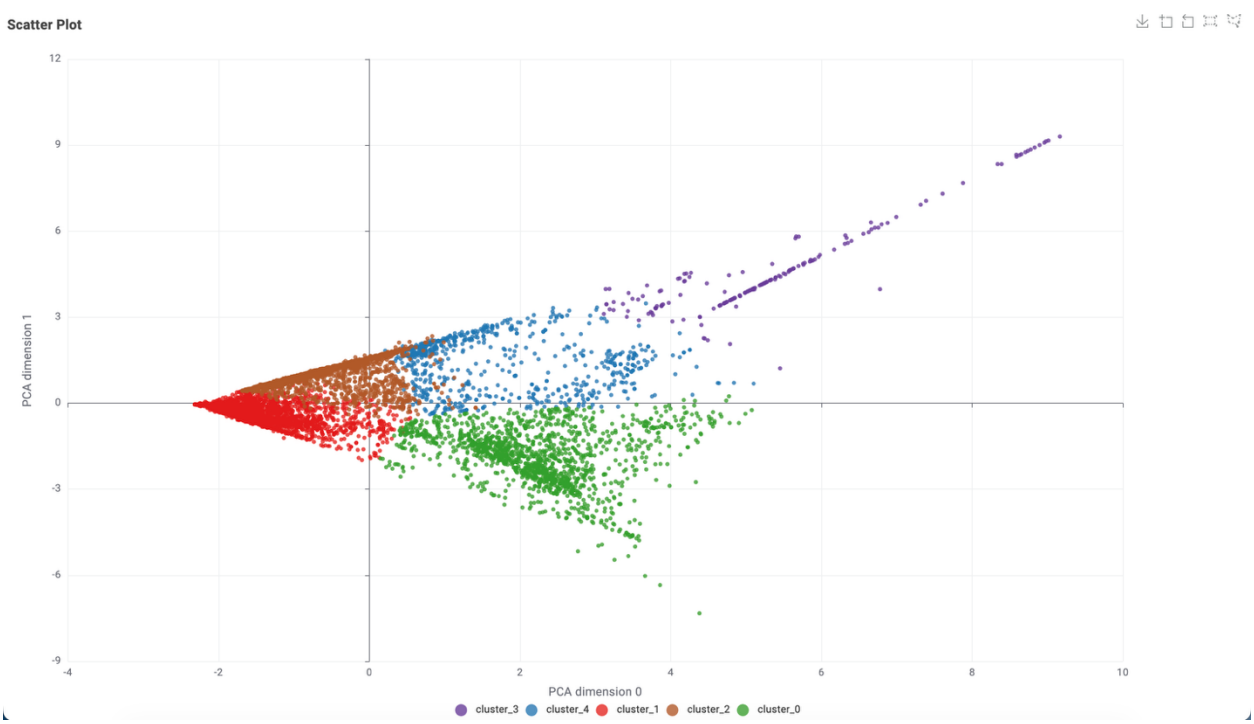
Name	Type	# Missing val...	# Unique valu...	Minimum	Maximum	25% Quantile	50% Quantile ...	75% Quantile	Mean	Mean Absolut...	Standard Devi...	Sum	10 most com...
Data.Carbohydra	Number (Float)	0	2955	-0.803	2.884	-0.785	-0.461	0.325	0	0.809	1	0	-0.803 (1708; 23
Data.Fiber	Number (Float)	0	235	-0.464	17.938	-0.464	-0.394	0.071	0	0.545	1	0	-0.464 (3418; 46
Data.Kilocalorie	Number (Float)	0	656	-1.28	3.975	-0.802	-0.225	0.649	0	0.792	1	0	3.87 (80; 1.08%)
Data.Protein	Number (Float)	0	2428	-1.031	7.393	-0.828	-0.338	0.742	-0	0.822	1	-0	-1.031 (298; 4.0
Data.Sugar Tota	Number (Float)	0	1654	-0.458	6.697	-0.458	-0.442	-0.125	0	0.625	1	0	-0.458 (3579; 48
Data.Water	Number (Float)	0	3741	-1.785	1.451	-0.788	0.295	0.802	0	0.845	1	0	-1.785 (101; 1.3
Data.Fat.Saturat	Number (Float)	0	3259	-0.499	14.164	-0.481	-0.321	0.098	-0	0.534	1	-0	-0.499 (629; 8.4
Data.Fat.Total Li	Number (Float)	0	2199	-0.619	5.441	-0.575	-0.36	0.165	0	0.623	1	0	-0.619 (291; 3.9
PCA dimension	Number (Float)	0	7071	-2.315	-0.619	-1.417	-0.651	1.263	0	1.491	1.823	0	-1.705 (7; 0.09%
PCA dimension	Number (Float)	0	7071	-7.326	9.3	-0.661	-0.121	0.851	-0	1.033	1.438	-0	-0.274 (7; 0.09%
Cluster	String	0	5	?	?	?	?	?	?	?	?	?	cluster_1 (2874;
Cluster (#1)	String	0	5	?	?	?	?	?	?	?	?	?	cluster_1 (2874;

5. PCA Cluster Visualization

The clusters were visualized using a scatter plot of PC1 versus PC2. Each point represents a product, and colors indicate cluster membership.

The plot shows that several clusters are clearly separated, while others show slight overlap. One cluster appears far from the others, suggesting extreme nutritional values, while another cluster is tightly grouped, indicating high similarity between items.

This visualization confirms that the clustering algorithm successfully identified meaningful structure in the data.



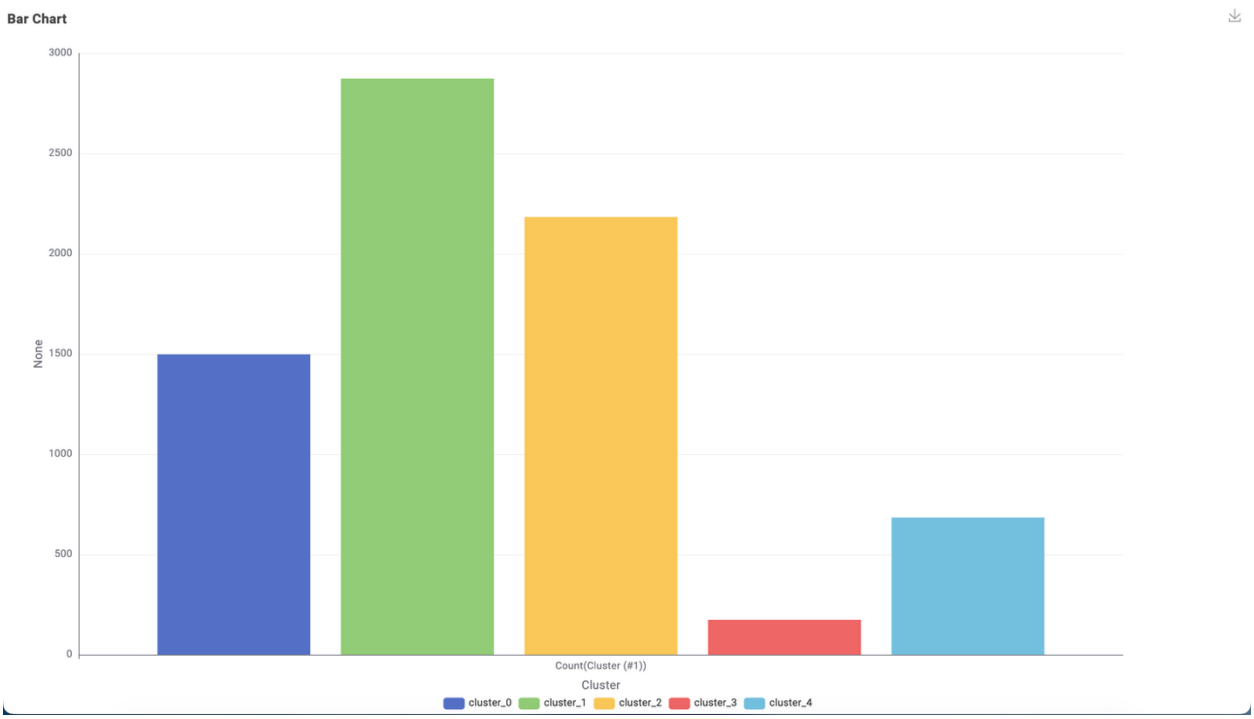
6. Cluster Distribution (GroupBy Analysis)

A GroupBy analysis was performed to count how many products belong to each cluster. This helps identify which nutritional profiles are most common and which are rare.

The results show that some clusters contain significantly more items than others. For example, cluster_1 and cluster_2 contain the majority of products, while cluster_3 is much smaller.

This suggests that certain nutritional profiles are overrepresented, while others represent niche or specialized products.

#	RowID	Cluster String	Count(Cluster (#1)) Number (Integer)
1	Row0	cluster_0	1498
2	Row1	cluster_1	2874
3	Row2	cluster_2	2183
4	Row3	cluster_3	174
5	Row4	cluster_4	684



7. Cluster Validation (Silhouette Coefficient)
To evaluate the quality of the clustering, silhouette coefficients were calculated. This metric measures how similar each data point is to its assigned cluster compared to other clusters.

Higher values indicate well-separated clusters, while lower values suggest overlap or possible misclassification.

The overall silhouette score of approximately 0.425 indicates moderate cluster separation. This suggests that most data points are appropriately grouped, although some overlap exists between clusters. This is expected given that many grocery products share similar nutritional characteristics.

#	RowID	Mean Silhouette Coefficient .00 Number (Float)
1	cluster	0.382
2	cluster	0.149
3	cluster	0.546
4	cluster	0.437
5	cluster	0.306
6	Overall	0.425

8. Cluster Interpretation (Nutritional Segments)
Each cluster represents a distinct nutritional profile:

Cluster 0 - Low-Calorie / High-Water Foods (Green)

- This cluster represents foods with high water content and low calories, such as fruits and vegetables. These items align well with wellness-focused products.

Cluster 1 - Low Energy / Light Foods (Red)

- This cluster contains lighter food options with relatively low fat and calorie content. These products are common and represent basic grocery items.

Cluster 2 - Balanced Nutritional Foods (Brown)

- This cluster includes products with a balanced nutritional composition, representing typical everyday foods without extreme values.

Cluster 4 - Moderate to High Energy Foods (Blue)

- This cluster includes foods with higher protein or carbohydrate content, resulting in moderate to high calorie levels.

Cluster 3 - High-Fat / High-Calorie Foods (Purple)

- This cluster represents extreme nutritional values, particularly high-fat and high-calorie items such as oils and processed fats. It is the smallest cluster, indicating a niche category.

9. Key Insights
The clustering results reveal clear patterns in the grocery inventory:

- Some nutritional profiles are highly overrepresented
- Certain product types form distinct and isolated clusters
- Extreme products (high fat or high calories) are clearly separated
- Balanced foods show more overlap

These findings demonstrate that nutritional composition is an effective way to segment grocery items into meaningful groups.

10. Conclusion

This project successfully developed an end-to-end clustering pipeline using PCA and k-Means to analyze grocery inventory data. The workflow was fully automated in KNIME, from preprocessing to visualization and validation.

The elbow method justified the selection of k = 5, and the clusters were validated using silhouette coefficients. The PCA visualization and cluster distribution analysis confirmed that meaningful patterns exist in the dataset.

Overall, this approach provides valuable insights into product segmentation and helps identify dominant product types and potential gaps in the inventory.

The KNIME workflow was fully automated, running from raw data input to final clustering results without manual intervention.